

# Primary VPT Chain: ProducerUA -> ProZorro -> Retail

## INTERPRETATION GUIDE

- ADF  $p > 0.05$  and KPSS  $p < 0.05$  -> likely I(1)-like; prefer differences/cointegration models.
- ADF  $p < 0.05$  and KPSS  $p > 0.05$  -> stationary; level models are more admissible.
- Ljung-Box  $p < 0.05$  -> autocorrelation; add lag structure.
- BP/White  $p < 0.05$  -> heteroskedasticity; use robust or HAC standard errors.
- JB  $p < 0.05$  -> non-normality; use robust inference and avoid strict normality assumptions.
- Stability flag=1 -> potential breaks/drift; use rolling or split-sample robustness.
- For retail transmission, compare no-promo vs promo-controlled estimates.

## RUN NOTES

standardized\_type=butter

retailer=silpo

promo\_variant=promo\_controlled

frequency=daily

Primary chain is strictly ProducerUA -> ProZorro -> Retail. Retail is estimated separately for silpo, novus, and silpo\_novus. Combined rule: daily median of available silpo and novus standardized\_type prices.

Table: PreTests

| series               | adf_level_p            | kpss_level_p | adf_diff1_p            | kpss_diff1_p | adf_diff2_p            | kpss_diff2_p        | integration_class | stability_flag | standardized_type | retailer | promo_variant    | cointegration_p       | pair_n_obs |
|----------------------|------------------------|--------------|------------------------|--------------|------------------------|---------------------|-------------------|----------------|-------------------|----------|------------------|-----------------------|------------|
| producer             | 0.08225840900720094    | 0.01         | 2.6818026212664985e-12 | 0.1          | 1.3006957963949979e-20 | 0.1                 | I(1)              | 1.0            | butter            | silpo    | promo_controlled | nan                   | nan        |
| prozorro             | 0.05640422694264767    | 0.01         | 3.518946761497874e-22  | 0.1          | 1.9269184182015793e-13 | 0.1                 | I(1)              | 1.0            | butter            | silpo    | promo_controlled | nan                   | nan        |
| retail               | 2.2989340931225115e-06 | 0.1          | 2.5595238291844733e-09 | 0.1          | 8.708607211858895e-05  | 0.04166666666666662 | I(0)              | 0.0            | butter            | silpo    | promo_controlled | nan                   | nan        |
| producer_to_prozorro | nan                    | nan          | nan                    | nan          | nan                    | nan                 | pair              | nan            | butter            | silpo    | promo_controlled | 0.46279135586174147   | 220.0      |
| prozorro_to_retail   | nan                    | nan          | nan                    | nan          | nan                    | nan                 | pair              | nan            | butter            | silpo    | promo_controlled | 8.244750384971287e-07 | 41.0       |
| producer_to_retail   | nan                    | nan          | nan                    | nan          | nan                    | nan                 | pair              | nan            | butter            | silpo    | promo_controlled | 0.9173535311228344    | 48.0       |

Table: ModelCoefficients

| standardized_type | retailer | promo_variant    | frequency | link                 | model_family | y_series | x_series | n_obs | sr_coef             | lr_coef              | ect_coef            | ect_pvalue             | asymmetry_short_p  | asymmetry_long_p    | vecm_rank | model_status | unreliable_flag | notes                        | primary_chain_doc            |
|-------------------|----------|------------------|-----------|----------------------|--------------|----------|----------|-------|---------------------|----------------------|---------------------|------------------------|--------------------|---------------------|-----------|--------------|-----------------|------------------------------|------------------------------|
| butter            | silpo    | promo_controlled | daily     | producer_to_prozorro | NARDL        | prozorro | producer | 218   | -0.2127764229425364 | 0.27405320585302084  | -0.8083252331134949 | 1.2523745815882856e-26 | 0.787381486847105  | 0.20112635504343956 | nan       | ok           | 0               | Wald short/long asymmetry... | Primary chain is strictly... |
| butter            | silpo    | promo_controlled | daily     | prozorro_to_retail   | ARDL         | retail   | prozorro | 41    | 0.20019436027147464 | -0.42577566334865624 | nan                 | nan                    | nan                | nan                 | nan       | unreliable   | 1               | bounds_lower_p=0.1096; bo... | Primary chain is strictly... |
| butter            | silpo    | promo_controlled | daily     | prozorro_to_retail   | NARDL        | retail   | prozorro | 39    | 0.04860976737983232 | -0.1264841571269484  | -0.8756967529857952 | 2.2144816825659353e-08 | 0.5941357916588494 | 0.5501601441286696  | nan       | unreliable   | 1               | Wald short/long asymmetry... | Primary chain is strictly... |

Table: ResidualDiagnostics

| model_family | link                 | y_series | x_series | ljungbox_p          | arch_p               | jb_p                   | unreliable_flag |
|--------------|----------------------|----------|----------|---------------------|----------------------|------------------------|-----------------|
| NARDL        | producer_to_prozorro | prozorro | producer | 0.30437404399912776 | 0.5287391650637225   | 4.2203813853019655e-84 | 0               |
| ARDL         | prozorro_to_retail   | retail   | prozorro | 0.01144894530682034 | 0.09344174279046094  | 0.6443904339547601     | 1               |
| NARDL        | prozorro_to_retail   | retail   | prozorro | 0.559171265338444   | 0.046665474894689134 | 0.8050949849331526     | 1               |

# Table: SeriesUsed

| date       | producer          | prozorro | retail | standardized_type | retailer | promo_variant    | combined_rule                |
|------------|-------------------|----------|--------|-------------------|----------|------------------|------------------------------|
| 2021-02-01 | 31.78268739915295 | nan      | nan    | butter            | silpo    | promo_controlled | daily_median_of_available... |
| 2021-02-02 | 31.74153855497394 | nan      | nan    | butter            | silpo    | promo_controlled | daily_median_of_available... |
| 2021-02-03 | 31.71205395008976 | nan      | nan    | butter            | silpo    | promo_controlled | daily_median_of_available... |
| 2021-02-04 | 31.69228954461795 | nan      | nan    | butter            | silpo    | promo_controlled | daily_median_of_available... |
| 2021-02-05 | 31.68030129867603 | nan      | nan    | butter            | silpo    | promo_controlled | daily_median_of_available... |
| 2021-02-06 | 31.67414517238153 | nan      | nan    | butter            | silpo    | promo_controlled | daily_median_of_available... |
| 2021-02-07 | 31.67187712585198 | nan      | nan    | butter            | silpo    | promo_controlled | daily_median_of_available... |
| 2021-02-08 | 31.6715531192049  | nan      | nan    | butter            | silpo    | promo_controlled | daily_median_of_available... |
| 2021-02-09 | 31.71501589339797 | nan      | nan    | butter            | silpo    | promo_controlled | daily_median_of_available... |
| 2021-02-10 | 31.82710410052748 | nan      | nan    | butter            | silpo    | promo_controlled | daily_median_of_available... |
| 2021-02-11 | 31.98036756741884 | nan      | nan    | butter            | silpo    | promo_controlled | daily_median_of_available... |
| 2021-02-12 | 32.14735612089748 | nan      | nan    | butter            | silpo    | promo_controlled | daily_median_of_available... |
| 2021-02-13 | 32.30061958778885 | nan      | nan    | butter            | silpo    | promo_controlled | daily_median_of_available... |
| 2021-02-14 | 32.41270779491835 | nan      | nan    | butter            | silpo    | promo_controlled | daily_median_of_available... |
| 2021-02-15 | 32.45617056911143 | nan      | nan    | butter            | silpo    | promo_controlled | daily_median_of_available... |
| 2021-02-16 | 32.45341900628176 | nan      | nan    | butter            | silpo    | promo_controlled | daily_median_of_available... |
| 2021-02-17 | 32.44522030312711 | nan      | nan    | butter            | silpo    | promo_controlled | daily_median_of_available... |
| 2021-02-18 | 32.43165843764893 | nan      | nan    | butter            | silpo    | promo_controlled | daily_median_of_available... |
| 2021-02-19 | 32.41281738784873 | nan      | nan    | butter            | silpo    | promo_controlled | daily_median_of_available... |
| 2021-02-20 | 32.38878113172797 | nan      | nan    | butter            | silpo    | promo_controlled | daily_median_of_available... |
| 2021-02-21 | 32.35963364728814 | nan      | nan    | butter            | silpo    | promo_controlled | daily_median_of_available... |
| 2021-02-22 | 32.32545891253073 | nan      | nan    | butter            | silpo    | promo_controlled | daily_median_of_available... |
| 2021-02-23 | 31.91148084818262 | nan      | nan    | butter            | silpo    | promo_controlled | daily_median_of_available... |
| 2021-02-24 | 30.90284056637734 | nan      | nan    | butter            | silpo    | promo_controlled | daily_median_of_available... |
| 2021-02-25 | 29.53949783222626 | nan      | nan    | butter            | silpo    | promo_controlled | daily_median_of_available... |
| 2021-02-26 | 28.06141241084077 | nan      | nan    | butter            | silpo    | promo_controlled | daily_median_of_available... |
| 2021-02-27 | 26.70854406733228 | nan      | nan    | butter            | silpo    | promo_controlled | daily_median_of_available... |
| 2021-02-28 | 25.72085256681216 | nan      | nan    | butter            | silpo    | promo_controlled | daily_median_of_available... |
| 2021-03-01 | 25.33829767439181 | nan      | nan    | butter            | silpo    | promo_controlled | daily_median_of_available... |
| 2021-03-02 | 25.36591783597624 | nan      | nan    | butter            | silpo    | promo_controlled | daily_median_of_available... |
| 2021-03-03 | 25.44026511447978 | nan      | nan    | butter            | silpo    | promo_controlled | daily_median_of_available... |
| 2021-03-04 | 25.54856970052784 | nan      | nan    | butter            | silpo    | promo_controlled | daily_median_of_available... |
| 2021-03-05 | 25.67806178474581 | nan      | nan    | butter            | silpo    | promo_controlled | daily_median_of_available... |
| 2021-03-06 | 25.81597155775909 | nan      | nan    | butter            | silpo    | promo_controlled | daily_median_of_available... |
| 2021-03-07 | 25.94952921019307 | nan      | nan    | butter            | silpo    | promo_controlled | daily_median_of_available... |
| 2021-03-08 | 26.06596493267315 | nan      | nan    | butter            | silpo    | promo_controlled | daily_median_of_available... |
| 2021-03-09 | 26.18214222460376 | nan      | nan    | butter            | silpo    | promo_controlled | daily_median_of_available... |
| 2021-03-10 | 26.31503339825227 | nan      | nan    | butter            | silpo    | promo_controlled | daily_median_of_available... |
| 2021-03-11 | 26.45203186353847 | nan      | nan    | butter            | silpo    | promo_controlled | daily_median_of_available... |
| 2021-03-12 | 26.58053103038216 | nan      | nan    | butter            | silpo    | promo_controlled | daily_median_of_available... |

## Table: LagProfile

|                      | pair | lag                  | corr |
|----------------------|------|----------------------|------|
| producer_to_prozorro | 1    | 0.07873337531291687  |      |
| producer_to_prozorro | 2    | 0.08283043430904553  |      |
| producer_to_prozorro | 3    | 0.08251800718380085  |      |
| producer_to_prozorro | 4    | 0.08422490303011267  |      |
| producer_to_prozorro | 5    | 0.0907848031216401   |      |
| producer_to_prozorro | 6    | 0.08185920607839353  |      |
| producer_to_prozorro | 7    | 0.08099913022176998  |      |
| producer_to_prozorro | 8    | 0.08418185337829556  |      |
| producer_to_prozorro | 9    | 0.08011971167494186  |      |
| producer_to_prozorro | 10   | 0.08071177659888301  |      |
| producer_to_prozorro | 11   | 0.0855520813606708   |      |
| producer_to_prozorro | 12   | 0.08891786146062777  |      |
| producer_to_prozorro | 13   | 0.0906935444584109   |      |
| producer_to_prozorro | 14   | 0.09336004953876562  |      |
| producer_to_prozorro | 15   | 0.09806722582073225  |      |
| producer_to_prozorro | 16   | 0.10411102969445078  |      |
| producer_to_prozorro | 17   | 0.11083781027515911  |      |
| producer_to_prozorro | 18   | 0.11554030327110705  |      |
| producer_to_prozorro | 19   | 0.11683243707057434  |      |
| producer_to_prozorro | 20   | 0.11514579002515733  |      |
| producer_to_prozorro | 21   | 0.11293989092313386  |      |
| producer_to_prozorro | 22   | 0.11348584116046659  |      |
| producer_to_prozorro | 23   | 0.11614446866624657  |      |
| producer_to_prozorro | 24   | 0.1193793796780393   |      |
| producer_to_prozorro | 25   | 0.12118388938915296  |      |
| producer_to_prozorro | 26   | 0.11984324455041497  |      |
| producer_to_prozorro | 27   | 0.11226973352035723  |      |
| producer_to_prozorro | 28   | 0.1007561552537856   |      |
| producer_to_prozorro | 29   | 0.09278063269691687  |      |
| producer_to_prozorro | 30   | 0.09120495228803738  |      |
| prozorro_to_retail   | 1    | 0.027215636352813634 |      |
| prozorro_to_retail   | 2    | -0.1306328650069521  |      |
| prozorro_to_retail   | 3    | 0.1914449373424407   |      |
| prozorro_to_retail   | 4    | 0.1048257656843792   |      |
| prozorro_to_retail   | 5    | -0.0807571026088193  |      |
| prozorro_to_retail   | 6    | -0.02993875084689255 |      |
| prozorro_to_retail   | 7    | -0.00117040461907769 |      |
| prozorro_to_retail   | 8    | -0.26096787569451063 |      |
| prozorro_to_retail   | 9    | -0.16753726563798146 |      |
| prozorro_to_retail   | 10   | 0.02498505897512193  |      |

Table: ModelEligibility

| standardized_type | retailer | promo_variant    | frequency | integration_producer | integration_prozorro | integration_retail | n_obs_producer | n_obs_prozorro | n_obs_retail | n_obs_pair_prod_prozorro | n_obs_pair_prozorro_retail | any_i2 | eligibility_note             |
|-------------------|----------|------------------|-----------|----------------------|----------------------|--------------------|----------------|----------------|--------------|--------------------------|----------------------------|--------|------------------------------|
| butter            | silpo    | promo_controlled | daily     | I(1)                 | I(1)                 | I(0)               | 1793           | 230            | 48           | 220                      | 41                         | 0      | Admissible for ARDL/ECM s... |

**Table: NARDL\_Multipliers**

| horizon | mult_pos             | mult_neg             | link                 |
|---------|----------------------|----------------------|----------------------|
| 0       | -0.5437076425441754  | -0.33093121960163896 | producer_to_prozorro |
| 1       | -0.48043463639437756 | -0.2924196897746266  | producer_to_prozorro |
| 2       | -0.48779791951561324 | -0.29690140029865086 | producer_to_prozorro |
| 3       | -0.4869410305889971  | -0.2963798492381926  | producer_to_prozorro |
| 4       | -0.4870407495079179  | -0.29644054381166113 | producer_to_prozorro |
| 5       | -0.4870291448967695  | -0.29643348058903957 | producer_to_prozorro |
| 6       | -0.48703049536267273 | -0.2964343025589613  | producer_to_prozorro |
| 7       | -0.4870303382046144  | -0.2964342069036792  | producer_to_prozorro |
| 8       | -0.4870303564936028  | -0.2964342180353919  | producer_to_prozorro |
| 9       | -0.48703035436525444 | -0.2964342167399587  | producer_to_prozorro |
| 10      | -0.4870303546129372  | -0.2964342168907124  | producer_to_prozorro |
| 11      | -0.4870303545841136  | -0.29643421687316873 | producer_to_prozorro |
| 12      | -0.48703035458746785 | -0.2964342168752103  | producer_to_prozorro |
| 13      | -0.4870303545870775  | -0.29643421687497273 | producer_to_prozorro |
| 14      | -0.48703035458712296 | -0.2964342168750004  | producer_to_prozorro |
| 15      | -0.48703035458711763 | -0.29643421687499716 | producer_to_prozorro |
| 16      | -0.4870303545871183  | -0.29643421687499755 | producer_to_prozorro |
| 17      | -0.4870303545871182  | -0.2964342168749975  | producer_to_prozorro |
| 18      | -0.4870303545871182  | -0.2964342168749975  | producer_to_prozorro |
| 19      | -0.4870303545871182  | -0.2964342168749975  | producer_to_prozorro |
| 20      | -0.4870303545871182  | -0.2964342168749975  | producer_to_prozorro |
| 0       | 0.06009522591878243  | 0.011485458538950107 | prozorro_to_retail   |
| 1       | 0.06495635205286528  | 0.012414521735102687 | prozorro_to_retail   |
| 2       | 0.06534957043136674  | 0.012489674017401043 | prozorro_to_retail   |
| 3       | 0.06538137801852381  | 0.012495753114665945 | prozorro_to_retail   |
| 4       | 0.06538395094658635  | 0.01249624485520164  | prozorro_to_retail   |
| 5       | 0.06538415907173531  | 0.012496284632284308 | prozorro_to_retail   |
| 6       | 0.06538417590705933  | 0.012496287849867952 | prozorro_to_retail   |
| 7       | 0.06538417726887523  | 0.012496288110139541 | prozorro_to_retail   |
| 8       | 0.06538417737903304  | 0.01249628813119301  | prozorro_to_retail   |
| 9       | 0.06538417738794375  | 0.012496288132896036 | prozorro_to_retail   |
| 10      | 0.06538417738866455  | 0.012496288133033794 | prozorro_to_retail   |
| 11      | 0.06538417738872285  | 0.012496288133044937 | prozorro_to_retail   |
| 12      | 0.06538417738872757  | 0.012496288133045838 | prozorro_to_retail   |
| 13      | 0.06538417738872794  | 0.01249628813304591  | prozorro_to_retail   |
| 14      | 0.06538417738872797  | 0.012496288133045916 | prozorro_to_retail   |
| 15      | 0.06538417738872798  | 0.012496288133045918 | prozorro_to_retail   |
| 16      | 0.06538417738872798  | 0.012496288133045918 | prozorro_to_retail   |
| 17      | 0.06538417738872798  | 0.012496288133045918 | prozorro_to_retail   |
| 18      | 0.06538417738872798  | 0.012496288133045918 | prozorro_to_retail   |

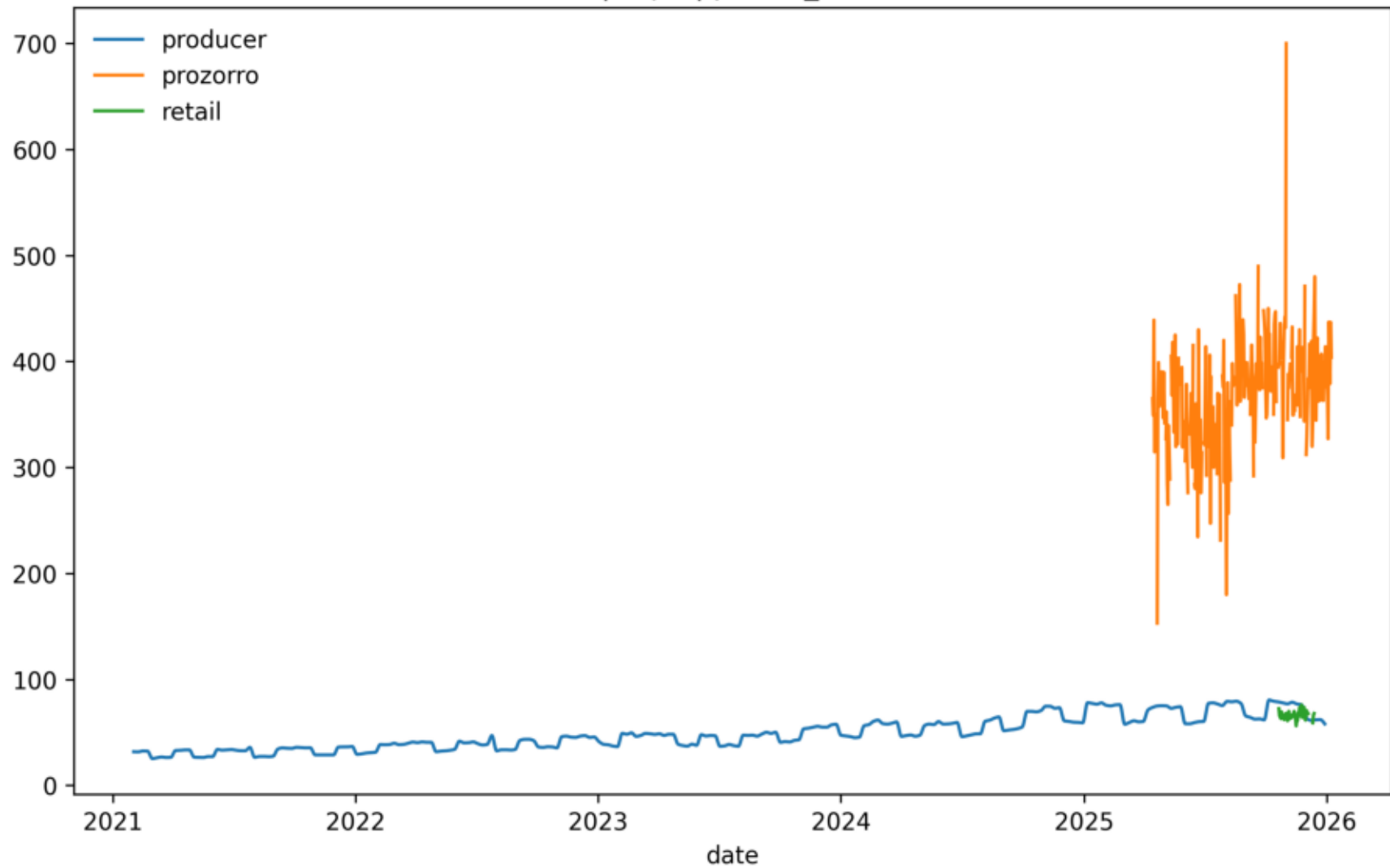


**Table: VECM\_IRF**

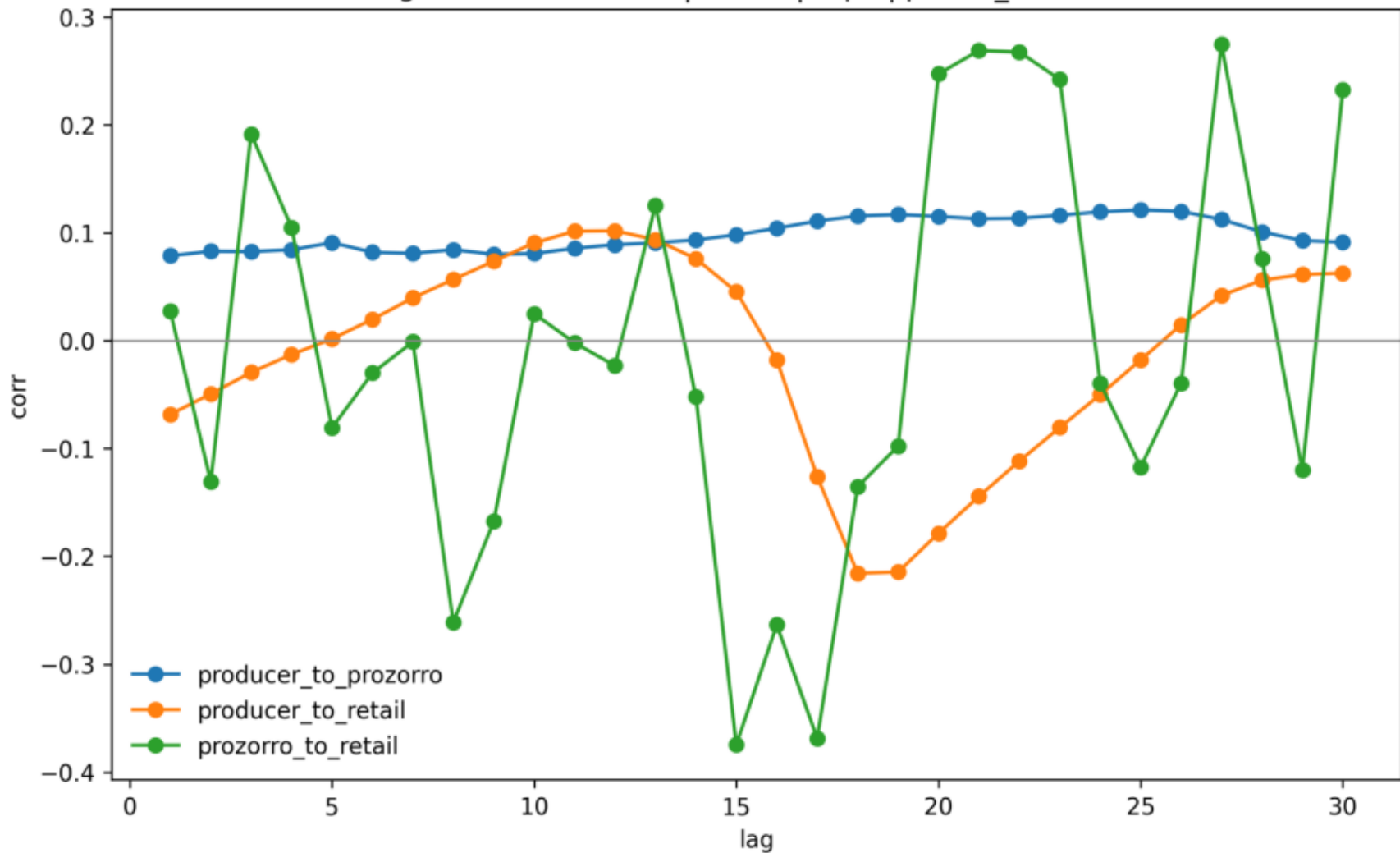
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butter | silpo | promo\_controlled



## Lag-Correlation Profile | butter | silpo | promo\_controlled



## NARDL Dynamic Multipliers | butter | silpo | promo\_controlled

